How the SnackAttack Map Icons Were Made

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Overview

The campus map now shows a colored pin for every vending machine that has GPS coordinates. Each pin's color and symbol show its machine type. Clicking a pin opens a popup with the building, payment options and directions.

The change is in PR #4 (`map-vending-icons` into `dev-sprint-2`). It touches two files:

- `index.html` : pin styles, icon builder, marker layer and popups
- `vending_machines.csv` : separate coordinates for the two Brinkley machines (#1 Drink, #2 Snack)

No new libraries, accounts or API keys were added.

Services and libraries used

Everything used is free and needs no account. Leaflet and the OpenStreetMap tiles were already on the page. Nominatim was used once, during development, and the live site never calls it.

Service	What it does here	Why this one	Used at
Leaflet 1.9.4	Draws the map, the pins and the popups	Already on the page; small, free, no API key; `L.divIcon` lets pins be plain HTML and CSS	Every page load
OpenStreetMap tiles	The map images behind the pins	Already in use; free; shows ETSU building outlines and names	Every page load

Service	What it does here	Why this one	Used at
Nominatim (OpenStreetMap search)	Returned the exact outline of the Millennium Center	Uses the same data as the map tiles, so its coordinates line up with what's drawn	Once, during development
`vending_machines.csv`	Holds each machine's type, GPS and details	Already the site's data source; pins read from it, so no second copy of the data	Every page load

The pin symbols are built-in emoji, styled with CSS, so there are no image files to host. Paid services such as Google Maps or Mapbox would need an API key and billing setup, and a school project doesn't need them.

Finding the machine locations

The two Brinkley machines are inside the Millennium Center, on its west side near the Niswonger Digital Media Center connector. They now sit about 5 m apart so their pins don't overlap.

1. Started from the survey coordinates in `miscFiles/VendingMachGenInfo`: 36°18′19.8″N 82°22′17.0″W, which is 36.305500, -82.371389 in decimal.
2. Looked up the Millennium Center on Nominatim. It returned the building outline, roughly 36.30501 to 36.30594 N and -82.37163 to -82.37065 W.
3. Matched the spots circled on the map screenshots against that outline. They fall about a third of the way along the west wall from the north end, which matches the survey point.
4. Both machines shared the same point, so their pins stacked into one. Each got its own spot at a circled location:

Machine	Type	Latitude	Longitude	Spot on screenshot
#1	Drink	36.305545	-82.371430	Lower-left circle
#2	Snack	36.305580	-82.371395	Upper-right circle

The first version had the two machines the other way around. They were swapped after review.

How the icons are built

Each pin is a small HTML element styled with CSS, not an image file. Leaflet places it on the map through `L.divIcon`.

- **Shape:** a 32 × 32 px box with three rounded corners and one square corner, rotated -45° so the square corner points down, like a map pin. A 2 px white border and a soft shadow help it stand out against the map.
- **Symbol:** an emoji inside the pin, rotated back +45° so it stays upright.
- **Anchor:** `iconAnchor: [16, 32]` puts the pin's tip, not its center, on the GPS point. `popupAnchor: [0, -30]` opens the popup just above the pin.

Machine type	Symbol	Pin color	CSS class
Drink	cup with straw	Blue `#1f6fd1`	`.machine-marker.drink`
Snack	chocolate bar	ETSU gold `#fdb515` (`--gold`)	`.machine-marker.snack`
Personal Care	lotion bottle	Purple `#8a3ffc`	`.machine-marker.personal-care`
Any other type	shopping cart	No color set yet	-

The CSS class comes from the type name, lowercased, with spaces turned into dashes ("Personal Care" becomes `personal-care`). A new type needs one CSS color rule and one entry in `markerEmoji`.

How icons connect to filters, search and popups

The pins always show the same machines as the list below the map. They're redrawn every time the list is.

1. `loadMachines()` reads `vending_machines.csv` and calls `render()`.

2. `render()` applies the type filter and building search, updates the list, then calls `renderMarkers(machines)` with the same filtered machines.
3. `renderMarkers()` clears the `markerLayer` group and adds one pin per machine.
4. `parseGPS()` turns the CSV's `"lat, lng"` text into numbers. Machines with blank or invalid GPS are skipped, so the 7 machines without coordinates yet simply don't show.
5. Each pin gets a popup with building and type in bold, then payment and directions. Missing values show "Not yet recorded".

All text in the popup goes through the existing `escapeHTML()` helper, so nothing typed into the CSV can break the page. Keeping all pins in one Leaflet `layerGroup` makes clearing and redrawing a single call.

Adding new machines and testing locally

To put a new machine on the map, fill in its `GPS` column in `vending_machines.csv`. No code changes are needed.

1. Find the spot on openstreetmap.org: right-click, then **Show address**, and copy the latitude and longitude.
2. Enter it in the CSV as `"36.305545, -82.371430"`: latitude first, quoted, comma and space between.
3. If two machines are at the same spot, move one a few meters (about 0.00003 to 0.00004 degrees) so both pins are visible.
4. Serve the folder with `python3 -m http.server 8000` and open http://localhost:8000. Opening `index.html` directly won't load the CSV.
5. Zoom in on the building and check that the pin tip sits where the machine is.

The map's starting view was moved slightly west, to 36.3035, -82.3700, so the Millennium Center pins show without panning. Machines spread across more of campus may call for a wider starting view.